

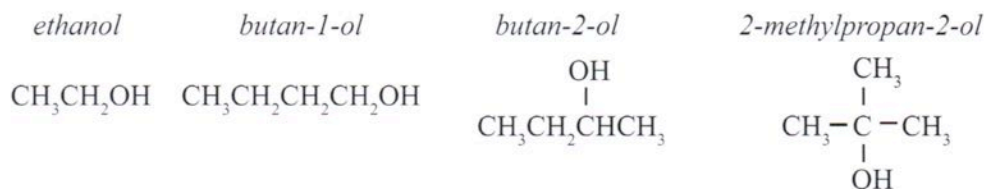
Experiment 31: Reactivity of alcohols



Notes

Alcoholic drinks contain the alcohol, ethanol. It is water soluble. It enters the bloodstream and with excessive consumption affects liver and brain function. Alcohols are organic molecules that contain the hydroxyl, (OH), functional group.

In this experiment you will investigate the alcohols, ethanol (C_2H_5OH) and three isomeric alcohols with formula C_4H_9OH , for their reactivity with the acidified solutions of oxidising agents potassium permanganate and sodium dichromate and the reducing agent sodium metal (optional). The structures of these alcohols are set out below.



Ethanol and butan-1-ol are primary alcohols because in both molecules the carbon atom bonded to the alcohol group has only one alkyl group bonded to it. Butan-2-ol is a secondary alcohol as the carbon atom bonded to the alcohol group has two alkyl groups bonded to it. 2-methylpropan-2-ol is a tertiary alcohol because it has three alkyl groups attached to the carbon atom to which the alcohol group is attached.

Equipment

beaker (250 mL)
thermometer (-10 to 110 °C)
14 large test tubes
Bunsen burner, tripod and gauze mat
dropper
metal tongs
graduated cylinder (10 mL)
sodium dichromate solution [$Na_2Cr_2O_7$] 0.1 mol L⁻¹ (5 mL)
sulfuric acid [H_2SO_4] 6 mol L⁻¹ (5 mL)
potassium permanganate solution [$KMnO_4$] 0.01 mol L⁻¹ (2 mL)
sodium metal (four pieces each of rice grain size)
6 mL of each alcohol:
ethanol [CH_3CH_2OH]
2-methylpropan-2-ol [$(CH_3)_3COH$]
butan-1-ol [$CH_3CH_2CH_2CH_2OH$]
butan-2-ol [$CH_3CH_2CHOHCH_3$]

Procedure

Part A: Reaction of alcohols with oxidising agents

Reaction with dichromate ion

1. Prepare a hot water bath by heating 100 mL of hot tap water in a 250 mL beaker to 80 °C. Turn the bunsen off.

SAFETY NOTE:

- Alcohols are flammable liquids and must be kept clear of naked flames. Make sure your Bunsen and all nearby Bunsens are turned off before you start using the alcohols.

2. Into four, separate, labelled test tubes place 2 mL of ethanol, butan-1-ol, butan-2-ol and 2-methylpropan-2-ol respectively.

SAFETY NOTE:

- **6 mol L⁻¹ sulfuric acid is corrosive and must be handled with care. If any H₂SO₄ comes in contact with your skin, immediately wash the affected area with plenty of water.**

3. In another test tube mix 4 mL of 0.1 mol L⁻¹ Na₂Cr₂O₇ and 2 mL of 6 mol L⁻¹ H₂SO₄.
4. Pour 1 mL of the acidified Na₂Cr₂O₇ solution into each of the test tubes containing the alcohols and heat them in the hot water bath for about 5 minutes. If the aqueous and alcoholic phases do not mix, shake the mixture occasionally.
5. Observe carefully for any evidence of reaction and note the relative rate of the reaction of each alcohol. Where a reaction has taken place carefully smell the contents of the tube. Write your observations in a suitable table.

Reaction with Permanganate Ion

6. Into four separate test tubes place 2 mL of ethanol, butan-1-ol, butan-2-ol and 2-methylpropan-2-ol respectively.
7. In another test tube mix 2 mL of 0.1 mol L⁻¹ KMnO₄ and 1 mL of 6 mol L⁻¹ H₂SO₄.
8. Add 3-5 drops of the acidified KMnO₄ to each of the test tubes containing the alcohols and shake gently.
9. Observe carefully for any evidence of reaction and note the relative rate of the reaction of each alcohol. Where a reaction has taken place, carefully smell the contents of the tube. Write your observations in a suitable table.

Procedure

Part B: Reaction of alcohols with sodium (optional)

SAFETY NOTE:

- **Sodium is a very reactive metal, particularly when it comes in contact with water. It must be handled with great care. Wear safety glasses. Use a spatula, plastic spoon or tweezers to handle the sodium. At the end of the experiment the residual sodium must be disposed of carefully by pouring the alcohol-sodium mixture into the beaker provided by your teacher.**

1. Into four, separate, dry test tubes place 2 mL of ethanol, butan-1-ol, butan-2-ol and 2-methylpropan-2-ol respectively.
2. Obtain from your teacher four small, freshly cut pieces of sodium.
3. Add one piece of sodium to each of the test tubes containing the alcohols. Observe carefully for any evidence of reaction and note the relative rate of the reaction of each alcohol.

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Processing of results, and questions

1. Which alcohols were oxidised by acidified $\text{Cr}_2\text{O}_7^{2-}$ solution and acidified MnO_4^- solution?
2. Write formulas for the products formed in the oxidation of the alcohols used.
3. Write balanced half equations then total equations for the oxidation of ethanol by acidified $\text{Cr}_2\text{O}_7^{2-}$ and acidified MnO_4^- solutions.
4. Write balanced half equations then total equations for the reaction of solutions of acidified $\text{Cr}_2\text{O}_7^{2-}$ and acidified MnO_4^- with primary, secondary and tertiary alcohols.
5. What was the observed order of reactivity of the $\text{C}_4\text{H}_9\text{OH}$ isomeric alcohols with sodium?
6. Write a general equation that represents the reaction of sodium with alcohols.
7. Write balanced half equations then a total equation for the reaction of sodium with alcohols.
8. There is a fourth isomeric alcohol with formula $\text{C}_4\text{H}_9\text{OH}$. Draw a structural formula for this isomer and write its systematic name. Predict how you would expect this isomer to react with
 - (a) acidified $\text{Na}_2\text{Cr}_2\text{O}_7$ solution
 - (b) metallic sodium.
9. A chemical plant involved in the production of solvents has received an order for a polar, water soluble solvent to be used in a commercial cleaning agent. The client has specified that the solvent should have low reactivity with a variety of common substances. As the plant chemist you are required to conduct preliminary tests to determine the relative reactivity of a number of polar, water soluble organic compounds. You decide that alcohols might fit the required criteria as they are able to hydrogen bond to water and generally make good solvents. Which of the alcohols tested in this experiment could be used in a commercial cleaning agent? Support your choice with evidence from this experiment.